

# Momentum and Collisions

SS ILP Physics (Intermediate), Week 5 Class 1



# Outline

- Momentum
- Impulse
- Momentum and Force

Reference:

Physics with Applications by Douglas Giancoli (7<sup>th</sup> Edition)

# Momentum and Impulse

If an object is moving, it has **momentum**.

The formal definition of momentum is that it's equal to an object's mass multiplied by that object's velocity. Mathematically, this is

$$\vec{P} = m\vec{v} \rightarrow P = mv$$

The units of momentum are  $\text{kg} \cdot \text{m/s}$  (kilogram meter per second).

As the momentum involves multiplying the object's velocity and the velocity has direction. Likewise, the momentum has direction.

# Momentum and Impulse

Impulse is an object's change in momentum. It is also equal to the **average force** acting on an object multiplied by the time interval over which that average force was applied. Mathematically, this is given by

$$\vec{I} = \Delta\vec{P} = \vec{F}_{\text{avg}}\Delta t \rightarrow I = \Delta P = F_{\text{avg}}\Delta t$$

The unit of impulse is the same as the momentum since it is, by definition, a change in momentum. An alternative description of the units (according to the formula) is N · s (Newton second).

# Momentum and Impulse

In Physics, units are a way of quantifying properties. Is  $\text{kg} \cdot \frac{\text{m}}{\text{s}} = \text{N} \cdot \text{s}$  ?

We start with the right-hand side and check to see if we get the left hand side of the equation.

Recall that N refers to Newton (unit of force) and force has units of  $\text{kg} \cdot \frac{\text{m}}{\text{s}^2}$ .

So,  $\text{N} \cdot \text{s} = \text{kg} \frac{\text{m}}{\text{s}^2} \cdot \text{s} = \text{kg} \cdot \frac{\text{m}}{\text{s}}$  which is the same as the left-hand side of the equation. So, the units for momentum and impulse are the same. This makes sense since impulse is the change of momentum.

The  $F_{\text{avg}}\Delta t$  definition of impulse explains why airbags are used in cars and why hitting someone with a pillow is less dangerous than hitting him or her with a cement block. The key is the  $\Delta t$  term.

# Summary

Momentum  $\vec{P} = m\vec{v} \rightarrow P = mv$

Impulse  $\vec{I} = \Delta\vec{P} = \vec{F}_{\text{avg}}\Delta t \rightarrow I = \Delta P = F_{\text{avg}}\Delta t$

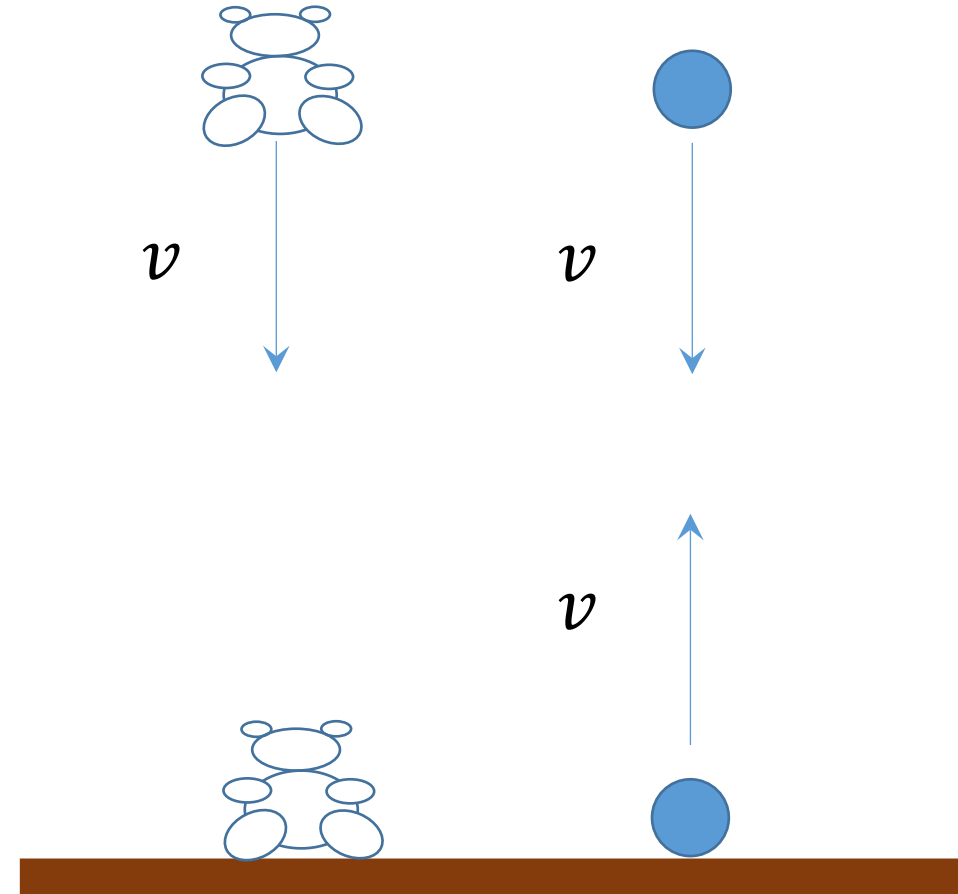
# Momentum and Impulse - Example

For the soft toy

$$\Delta P = m(0 - (-v)) = mv$$

For the steel ball

$$\Delta P = m(v - (-v)) = 2mv$$



From [https://web.njit.edu/~binchen/phys111/LectureNotes/Physics111\\_lecture08.pdf](https://web.njit.edu/~binchen/phys111/LectureNotes/Physics111_lecture08.pdf)

# Momentum and Impulse - Example

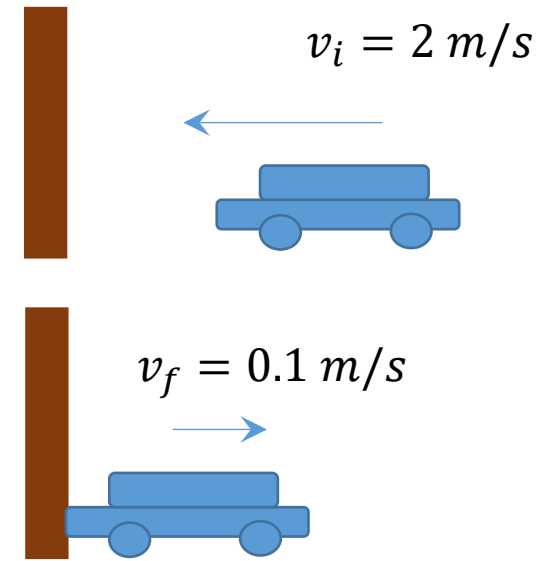
A toy car has a mass of 20g, colliding a wall with a collision time of 0.1s. Find

- (a) the impulse experienced by the car due to the collision, and
- (b) the magnitude and direction of the average force acting on the car.

$$I = P_f - P_i = mv_f - mv_i = 0.02[0.1 - (-2)] = 0.042 \text{ kg} \cdot \text{m/s}$$

$$F_{avg} = \frac{\Delta P}{\Delta t} = \frac{I}{\Delta t} = \frac{0.042}{0.1} = 0.42 \text{ N}$$

to the right!





## Concept Question 1:

A ball of mass  $M$  is caught by someone wearing a baseball glove. The ball is in contact with the glove for a time  $t$  ; the initial velocity of the ball (just before the catcher touches it) is  $v_0$ .

If the time of the ball's collision with the glove is doubled, what happens to the average force necessary to catch the ball?

- (A) It doesn't change.
- (B) It is cut in half.
- (C) It is cut to one fourth of the original force.
- (D) It quadruples.
- (E) It doubles.

## Concept Question 1: Solution

**Answer is B.**

Impulse is force times the time interval of collision and is also equal to an object's change in momentum.

Solve for force and we get  $F_{\text{avg},1} = \Delta P / \Delta t_1 = \Delta P / t$ .

Because the ball still has the same mass, and still changes from speed  $v_0$  to a speed of zero, the ball's momentum change is the same, regardless of the collision time.

If the collision time is doubled, then the new force

$$F_{\text{avg},2} = \frac{\Delta P}{\Delta t_2} = \frac{\Delta P}{2t} = \frac{1}{2} \left( \frac{\Delta P}{t} \right) = \frac{1}{2} F_{\text{avg},1} .$$

So, the force has now been cut in half.

## Concept Question 2:

A ball of mass  $M$  is caught by someone wearing a baseball glove. The ball is in contact with the glove for a time  $t$  ; the initial velocity of the ball (just before the catcher touches it) is  $v_0$ .

If the time of collision remains  $t$ , but the initial velocity is doubled, what happens to the average force necessary to catch the ball?

- (A) It doesn't change.
- (B) It is cut in half.
- (C) It is cut to one fourth of the original force.
- (D) It quadruples.
- (E) It doubles.

## Concept Question 2: Solution

**Answer is E.**

Still use  $F_{\text{avg},1} = \Delta P / \Delta t$ , but this time it is the numerator that changes.

The ball still is brought to rest by the glove, and the mass of the ball is still the same.

However, the doubled velocity upon reaching the glove doubles the momentum change.

$$\text{The } F_{\text{avg},2} = \frac{2 \cdot \Delta P}{\Delta t} = 2 \left( \frac{\Delta P}{\Delta t} \right) = 2 F_{\text{avg},1}$$

So the force doubles.

## Case Problem 1:

For a top player, a tennis ball may leave the racket on the serve with a speed of 55 m/s. If the ball has a mass of 0.060 kg and is in contact with the racket for about 4 ms (milliseconds), estimate the average force on the ball. Would this force be large enough to lift a 60-kg person?

## Case Problem 1: Solution

For a top player, a tennis ball may leave the racket on the serve with a speed of 55 m/s. If the ball has a mass of 0.060 kg and is in contact with the racket for about 4 ms (milliseconds), estimate the average force on the ball. Would this force be large enough to lift a 60-kg person?

Solution: The tennis ball is hit when its initial velocity  $v_i$  is very nearly zero at the top of the throw, so we set the  $v_i = 0$ . We let the final velocity  $v_f$  be the speed the ball leaves the tennis racket and assume that it is in the horizontal direction.

Recall that  $\Delta P = F_{\text{avg}}\Delta t$ , where  $\Delta P$  refers to the difference between the initial and final momenta,  $F_{\text{avg}}$  is the average force and  $\Delta t$  is the time interval between the initial and final momenta.

## Case Problem 1: Solution

The initial momenta is  $P_i = mv_i = (0.06)(0) = 0$  and the final momenta is  $P_f = mv_f = (0.06)(55) = 3.3 \text{ kg m/s}$ .

So  $\Delta P = 3.3 - 0 = 3.3 \text{ kg m/s}$ .

We ignore all other forces on the ball during this brief time interval, such as gravity, in comparison to the force exerted by the tennis racket. More to be said later.

Convert 4 milliseconds to 0.004 seconds. The force exerted on the ball by the racket is

$$F_{\text{avg}} = \frac{\Delta P}{\Delta t} = \frac{3.3}{0.004} = 825 \text{ N}$$

This is a large force but to answer the question whether this force can lift a 60 kg person, we need to translate the mass of this person into a weight since weight has the same units as force.

## Case Problem 1: Solution

Weight of a 60-kg person = force of gravity on 60 kg person =  $mg = 60 \text{ kg} \times 9.8 \text{ m/s}^2 = 588 \text{ N}$ . Since the force exerted on the ball by the racket is greater than the weight, this force is able to lift a 60 kg person.

Earlier, we said that we will ignore other forces on the ball. This has to be investigated a little further.

The force of gravity acting on the ball is  $mg = 0.06 \text{ kg} \times 9.8 \text{ m/s}^2 = 0.59 \text{ N}$ , which justifies our ignoring it compared to the enormous force the racket exerts.

Extra comments: High-speed photography and radar can give us an estimate of the contact time and the velocity of the ball leaving the racket. But a direct measurement of the force is often not practical.

Our calculation shows a handy technique for determining an unknown force in the real world.



## Case Problem 2:

A man jumps off the roof of a building, 3 m above the ground. His mass is 70 kg. He estimates (while in free-fall) that if he lands stiff-legged, it will take him 3 ms (milliseconds) to come to rest. However, if he bends his knees upon impact, it will take him 100 ms to come to rest. Which way will he choose to land, and why?

## Case Problem 2: Solution

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Solution: This is a multistep problem. We start by calculating the man's velocity the instant before he hits the ground. That's a kinematics problem, so we start by choosing a positive direction—we'll choose "down" to be positive—and by writing out our table of variables.

|            |                     |
|------------|---------------------|
| $v_0$      | 0                   |
| $v_f$      | ?                   |
| $\Delta x$ | 3.0 m               |
| $a$        | 10 m/s <sup>2</sup> |
| $t$        | ?                   |

## Case Problem 2: Solution

We have three variables with known values, so we can solve for the other two. We don't care about time for the moment, so we will just first solve for  $v_f$ .

$$v_f^2 = v_0^2 + 2a \Delta x = 0 + 2(10)(3) \quad \rightarrow \quad v_f = 7.7 \text{ m/s}$$

Now we can solve for the man's momentum the instant before he hits the ground.

$$P = mv_f = (70)(7.7) = 540 \text{ kg m/s}$$

Once he hits the ground, the man quickly comes to rest. So the momentum changes from 540 kg m/s to 0. So the change of momentum is negative,

$$\Delta P = 0 - 540 = -540 = F_{\text{avg}} \Delta t$$

## Case Problem 2: Solution

If the man does not bend his knees (stiff-legged), then he takes 3 milliseconds (ms) = 0.003 s to come to rest.

$$-540 = F_{\text{avg},1} \Delta t_1 \quad \rightarrow \quad F_{\text{avg},1} = -\frac{540}{0.003} = -180,000 \text{ N}$$

What does the negative sign in the answer mean? As we said previously, we will choose “down” as the positive direction. This means the force of -180,000 N is in the negative direction, i.e. directed upwards.

$$\text{If the man bent his knees, then } -540 = F_{\text{avg},2} \Delta t_2 \rightarrow F_{\text{avg},2} = -\frac{540}{0.10} = -5,400 \text{ N}$$

If he bends his knees, he allows for his momentum to change more slowly, and as a result, the ground exerts a lot less force on him than had he landed stiff-legged.

## Case Problem 2: Solution

Extra comments: More to the point, hundreds of thousands of Newtons applied to a persons' legs will cause major damage.

This is the equivalent of almost (approximation) 20,000 kg sitting on his legs.

So we would assume that the man would bend his knees upon landing, reducing the force on his legs by a factor of 30!

So when you jump, remember to bend your knees on the way down!

# Momentum and Force

From the definition of impulse we just covered,  $\Delta P = F_{\text{avg}}\Delta t$ .

As the time interval is  $\Delta t$  (assuming non-zero for now), we can divide this term on both sides of above equation,

$$\Delta P / \Delta t = F_{\text{avg}}$$

So, the force applied is the change in momentum divided by the time interval. Let us explore this further.

If the time interval became smaller, the change in momentum is also expected to get smaller too.

# Momentum and Force

This should make you recall the ideas of differential, i.e. when  $\Delta t \rightarrow 0$ ,

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta P}{\Delta t} = \frac{dP}{dt} = F$$

When we write

$$\frac{dP}{dt} = F \rightarrow \vec{F} = \frac{d\vec{P}}{dt}$$

we are referring to an instantaneous momentum  $P$  and instantaneous force  $F$ .

By instantaneous, we are referring to the value of the momentum and force at an instant in time.

In other words,  $P(t)$  and  $F(t)$  where  $t$  is a particular value of time.

# Momentum and Force

- If the momentum  $P(t)$  is a constant, i.e.  $P(t) = A$ , then the force  $F(t) = 0$ . Since  $\frac{d}{dt}A = 0$ .

Physically, this makes sense because momentum is mass multiplied by velocity. So, if momentum is a constant, then velocity is constant. There is no acceleration and by Newton's second law, the force is zero.

- If the momentum  $P(t)$  is a linear function in time, i.e.  $P(t) = At + B$ , then the force  $F(t) = A$ , i.e. a constant.

Physically this makes sense too because mass is assumed constant, so only the velocity is changing with time.



# Summary

Momentum  $\vec{P} = m\vec{v} \rightarrow P = mv$

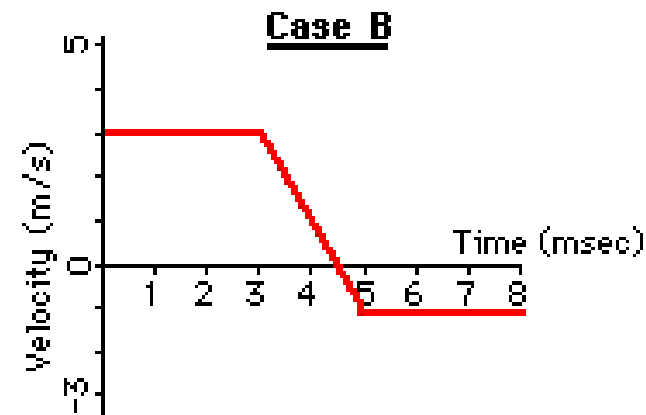
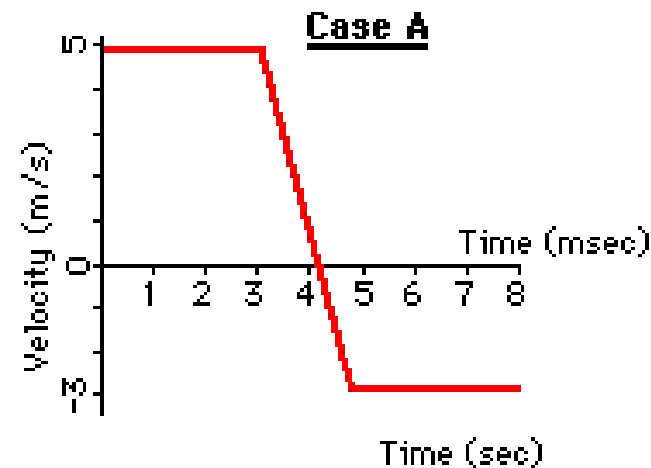
Impulse  $\vec{I} = \Delta\vec{P} = \vec{F}_{\text{avg}}\Delta t \rightarrow I = \Delta P = F_{\text{avg}}\Delta t$

Momentum and Force

$$\vec{F} = \frac{d\vec{P}}{dt} \rightarrow F = \frac{dP}{dt}$$

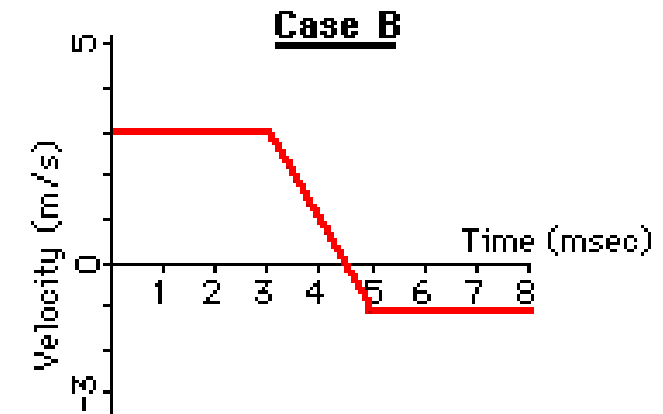
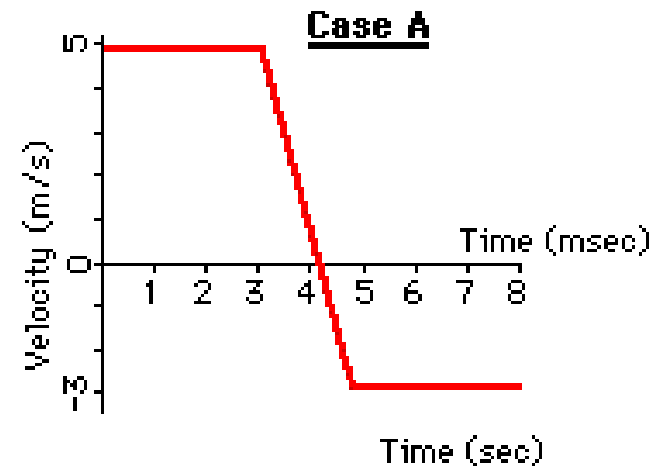
## Case Problem 3:

For the two cases A and B, determine the momentum and force as functions of time. Assume that the mass of the object featured in the cases A and B is 2 kg and 4 kg respectively.



## Case Problem 3: Solution

For the two cases A and B, determine the momentum and force as functions of time. Assume that the mass of the object featured in the cases A and B is 2 kg and 4 kg respectively.

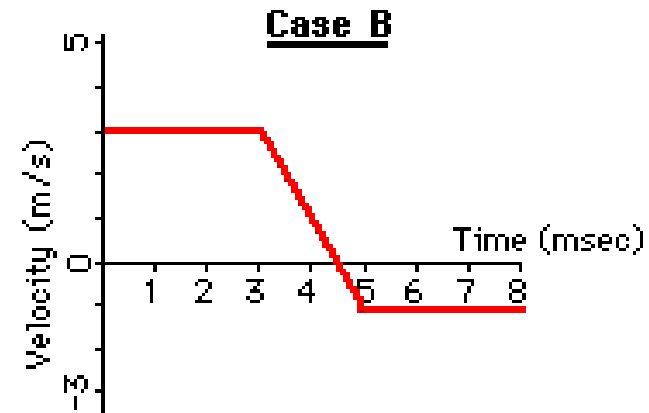
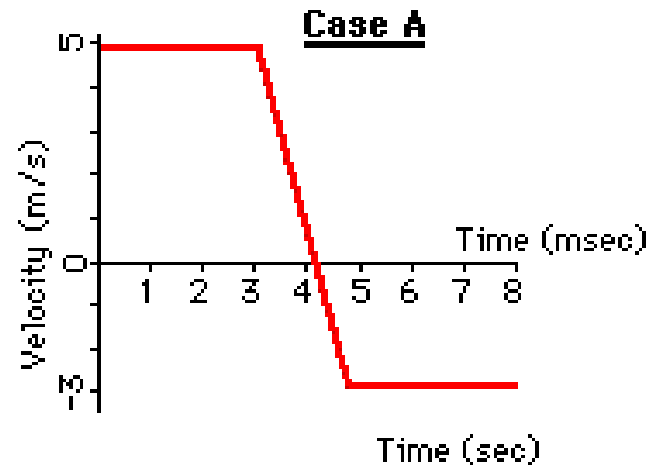


Solution: First, we want to write the equations for the velocity curve for both cases and then we evaluate the momentum functions, followed by the force functions. Starting with case A, we have

$$\text{Case A: } v_A(t) = \begin{cases} 5 & , \quad 0 \leq t < 3 \\ -4t + 17 & , \quad 3 \leq t \leq 5 \\ -3 & , \quad 5 < t \leq 8 \end{cases}$$

To obtain the function from  $t = 3$  to  $t = 5$ , we have to identify two points (3,5) and (5,-3). The general formula is given by  $y = mt + c$  where  $m$  is the slope and  $c$  is the intercept value on the velocity axis. Slope is  $m = \frac{\Delta v}{\Delta t} = \frac{-3-5}{5-3} = -4$  and  $5 = -4(3) + c \rightarrow c = 17$

# Case Problem 3: Solution

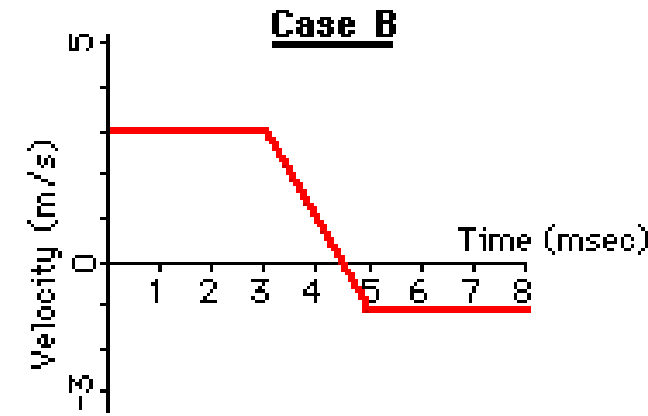
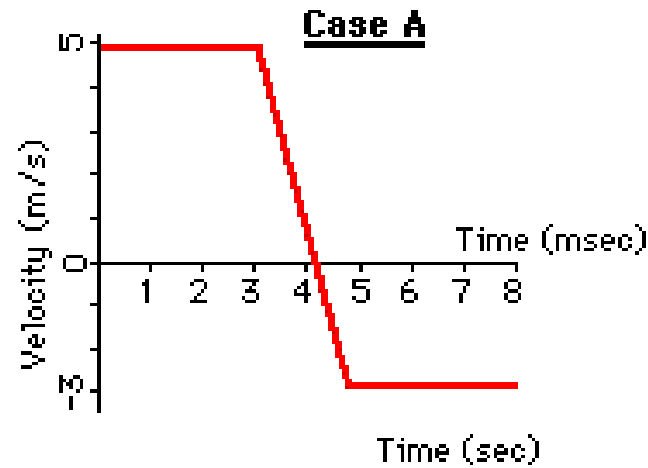


$$\rightarrow P_A(t) = m_A v_A(t) = \begin{cases} 10 & , \quad 0 \leq t < 3 \\ -8t + 34 & , \quad 3 \leq t \leq 5 \\ -6 & , \quad 5 < t \leq 8 \end{cases}$$

$$\rightarrow F_A(t) = dP_A/dt = \begin{cases} 0 & , \quad 0 < t < 3 \\ -8 & , \quad 3 < t < 5 \\ 0 & , \quad 5 < t < 8 \end{cases}$$

Recall that the derivative of a linear function is a constant and the derivative of a constant is zero.

# Case Problem 3: Solution



$$\text{Case B: } v_B(t) = \begin{cases} 3 & , \quad 0 \leq t < 3 \\ -2t + 9 & , \quad 3 \leq t \leq 5 \\ -1 & , \quad 5 < t \leq 8 \end{cases}$$

To obtain the function from  $t = 3$  to  $t = 5$ , we have to identify two points (3,3) and (5,-1). The general formula is given by  $y = mt + c$  where  $m$  is the slope and  $c$  is the intercept value on the velocity axis. Slope is  $m = \frac{\Delta v}{\Delta t} = \frac{-1-3}{5-3} = -2$  and  $3 = -2(3) + c \rightarrow c = 9$

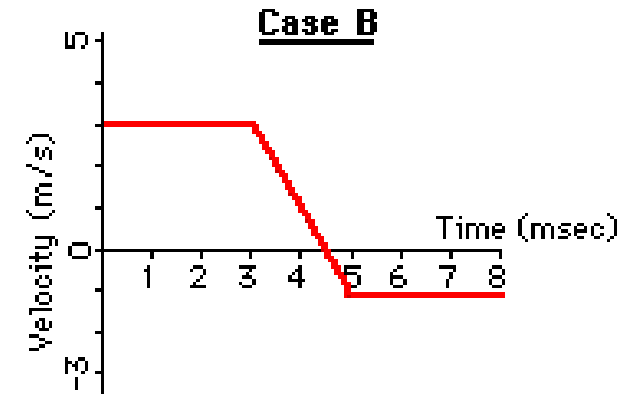
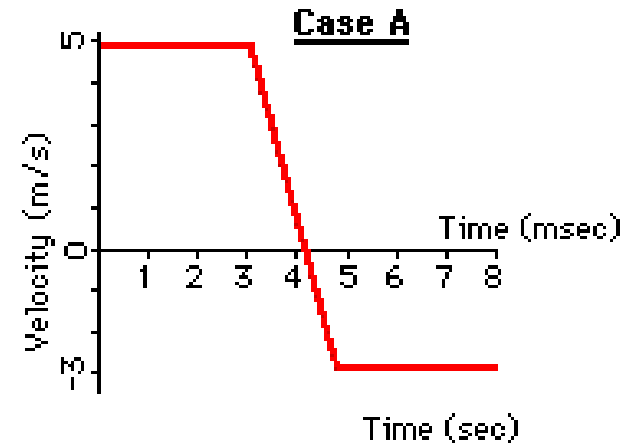
$$\rightarrow P_B(t) = m_B v_B(t) = \begin{cases} 12 & , \quad 0 \leq t < 3 \\ -8t + 36 & , \quad 3 \leq t \leq 5 \\ -4 & , \quad 5 < t \leq 8 \end{cases} \quad \rightarrow F_B(t) = dP_B/dt = \begin{cases} 0 & , \quad 0 < t < 3 \\ -8 & , \quad 3 < t < 5 \\ 0 & , \quad 5 < t < 8 \end{cases}$$

# Off-class practice question(s)

## Concept Question 3:

Of the two cases A and B, which has the

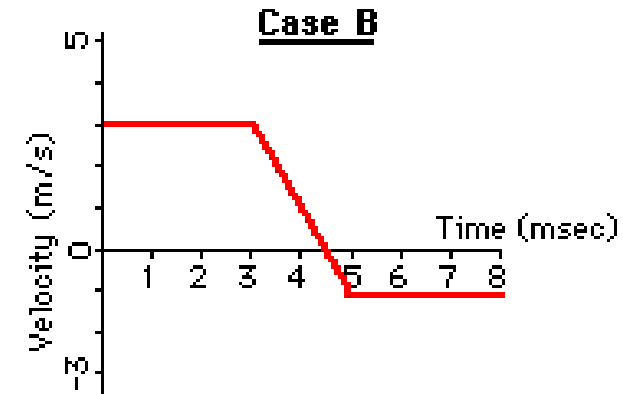
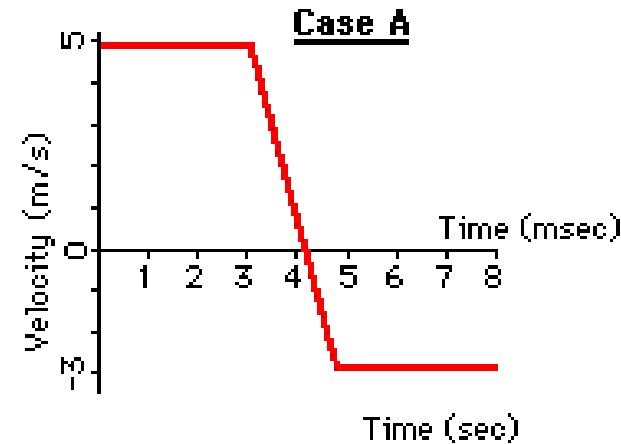
- (i) greatest velocity change,
- (ii) greatest acceleration,
- (iii) greatest change of momentum
- (iv) greatest impulse?



## Concept Question 3: Solution

Of the two cases A and B, which has the

- (i) greatest velocity change,
- (ii) greatest acceleration,
- (iii) greatest change of momentum
- (iv) greatest impulse?



(i). The velocity change is greatest in case A.

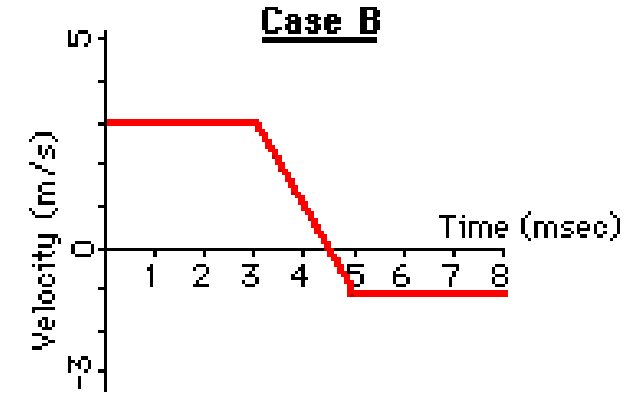
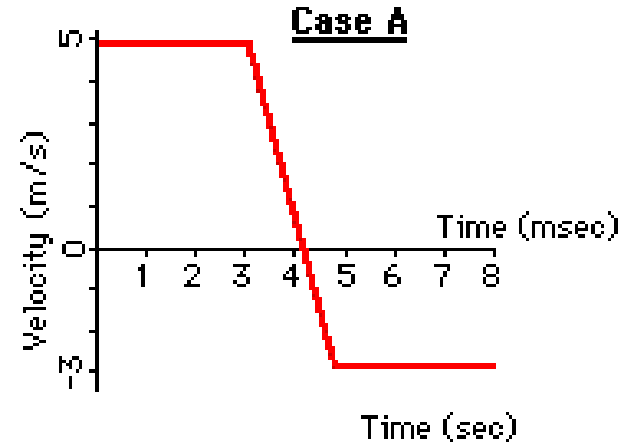
The  $v$  changes from +5 m/s to -3 m/s. This is a change of 8 m/s (-) and is greater than in case B (-4 m/s).



## Concept Question 3: Solution

Of the two cases A and B, which has the

- (i) greatest velocity change,
- (ii) greatest acceleration,
- (iii) greatest change of momentum and
- (iv) greatest impulse?



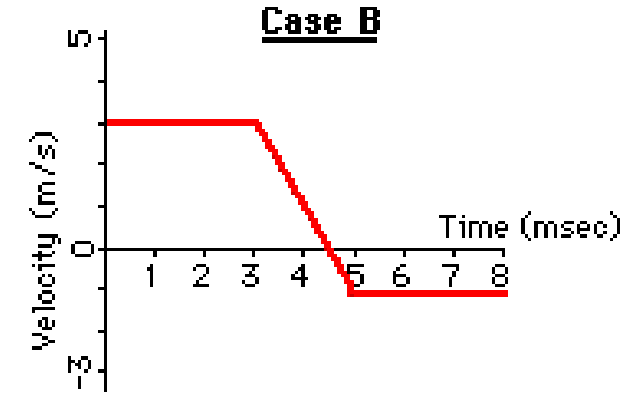
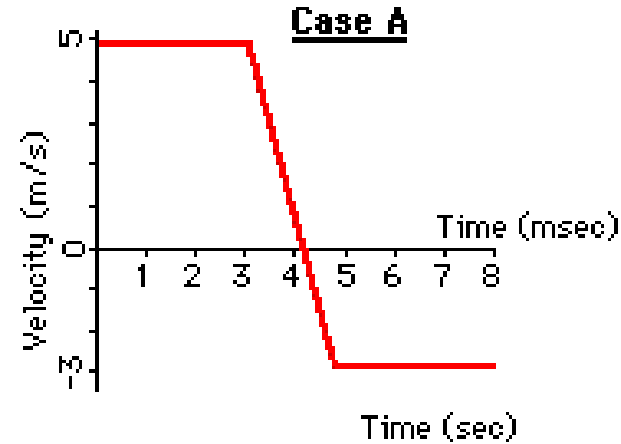
(ii). The acceleration is greatest in case A.

Acceleration depends on velocity change and the velocity change is greatest in case A (as stated above). Furthermore, the magnitude of the non-zero slope (gradient) of the velocity curve in case A is greater than in case B.

## Concept Question 3: Solution

Of the two cases A and B, which has the

- (i) greatest velocity change,
- (ii) greatest acceleration,
- (iii) greatest change of momentum
- (iv) greatest impulse?



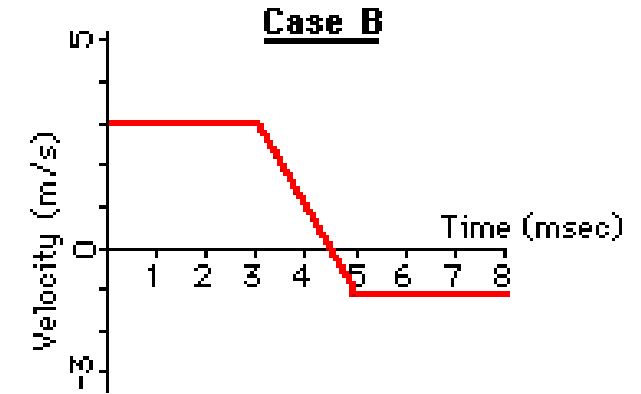
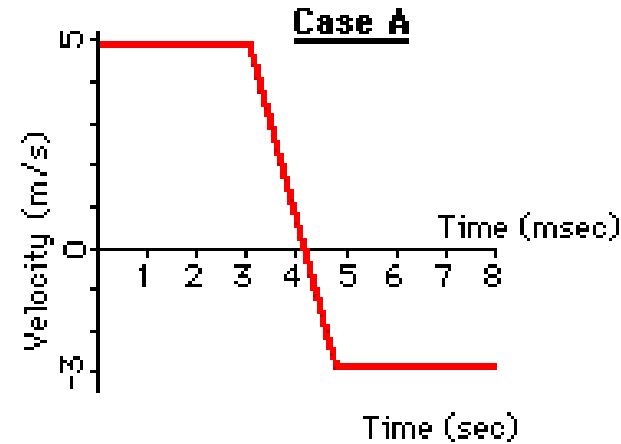
(iii). The momentum change is greatest in case A.

Momentum change depends on velocity change and the velocity change is greatest in case A (as stated above).

## Concept Question 3: Solution

Of the two cases A and B, which has the

- (i) greatest velocity change,
- (ii) greatest acceleration,
- (iii) greatest change of momentum
- (iv) greatest impulse?



iv. The impulse is greatest in case A.

Impulse equals momentum change and the momentum change is greatest in case A (as stated above).